DIAR ABDLKARIM

Postdoctoral Research Scientist · School of Psychology, University of Birmingham

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PROFESSIONAL SUMMARY

I am a postdoctoral research scientist at the University of Birmingham, currently working on non-invasive brain–computer interface (BCI) systems for decoding naturalistic human movement within an ARIA-funded programme. I draw on expertise in neural engineering, real-time signal processing, motion capture and neurophysiology to generate scientific results from first principles. **Alongside my own research, I have built a sustained track record of strengthening research culture across the University**, conceiving and delivering cross-disciplinary training programmes, establishing shared experimental infrastructure, and seeding collaborations that connect Psychology, Computer Science, Engineering, Sport & Exercise Sciences and the Arts. A recurring theme of my work is bringing advanced methods (motion capture, immersive XR and kinematic analysis) to research communities that would not otherwise have access to them, raising technical capability and enabling new lines of inquiry that have produced peer-reviewed publications, student theses and follow-on funding.

Areas of expertise: Motion capture & 3D kinematic analysis · immersive XR / VR & game-engine development · non-invasive BCI & neural signal processing · haptics & multisensory feedback · experimental design with human participants · computer vision · rehabilitation technology.

ENHANCING RESEARCH CULTURE

A consistent focus of my work at Birmingham has been to enhance research culture: equipping colleagues across disciplines with advanced methods, building durable shared infrastructure, and creating inclusive, collaborative communities of practice. The contributions below span training, infrastructure, cross-departmental collaboration and mentorship, and have measurably raised methodological capability across multiple Schools.

Cross-disciplinary research training I conceived, designed and delivered

- **Motion Capture Crash Course.** Identified an unmet need for hands-on motion-capture skills across the University and independently conceptualised, designed and delivered a **3-day intensive, hands-on course** covering the full pipeline, system setup, calibration, data capture, processing and kinematic analysis for research, rehabilitation and HCI. The programme has trained **60+ faculty, postdocs, PhD students and technical staff** drawn from Psychology, Computer Science, the Arts and beyond. Our post-course survey returned **98% satisfaction**, with attendees reporting that they left confident to apply the technology independently in their own research. The course has lifted the baseline technical capability of my department and partner Schools and directly catalysed new interdisciplinary projects.
- **XR & Game-Engine Crash Course for Scientists.** Identified the need and created what is, to my knowledge, the **only dedicated course in the world** on virtual reality and game-engine development built specifically for scientists. It has run for **3+ years, twice yearly**, each cohort a full **five-day (one-week) intensive**, hands-on and real-world applicable. To date it has reached **90+ participants**, professors, postdocs, PhD and master's students, with a **95% satisfaction rate** and attendees leaving confident to use game engines and VR in their research. Many alumni have since published work or progressed to PhD study on the strength of skills gained in the course.
- **Research Methodology Course for non-Psychology master's students.** Recognised that master's students outside Psychology often lack training in working with human participants and running scientific studies. I conceived, designed and delivered a **one-day intensive, hands-on course**, rich in

real-world examples, that takes participants from no prior experience to being able to design and run their own experiments with human participants, widening access to rigorous empirical methods beyond the discipline.

Cross-departmental infrastructure & collaboration I enabled

- **Sign-language research (Psychology × Computer Science).** Introduced motion-capture and kinematic-analysis methods, established in Computer Science, to a **British Sign Language research group in Psychology**, a novel methodological approach in the field. I set up the capture system and trained the team, enabling the kinematic study of sign-language learning and production (including a PhD thesis project). Outputs to date include a peer-reviewed article in *Scientific Reports* and presentations and posters at international conferences, with further papers in preparation for high-impact journals.
- **Gait & movement-science motion-capture laboratory (Sport, Exercise & Rehabilitation Sciences).** Helped senior movement-science researchers specify and establish a **motion-capture laboratory for gait analysis**, durable infrastructure that will support studies across the School for years to come.
- **Extreme Robotics Lab (ERL).** Supported the ERL in setting up motion-capture systems for **collaborative human–robot work** on complex object manipulation and grasping under uncertain, undefined conditions, enabling new research at the human–robot interface.

Research community & mentorship

- **BhamXR.** Helped establish a **cross-college community** for researchers working on extended-reality applications, connecting Psychology, Computer Science and other Schools and creating a forum for shared learning and collaboration.
- **Supervision & mentoring.** Supervised **30+ undergraduate and master's projects** (Psychology, Computational Neuroscience & Cognitive Robotics, and Computer Science) and **2 PhD students**, mentoring the next generation of researchers in advanced methods and in open, collaborative practice.

Taken together, this work has raised the methodological capability of researchers across several Schools, seeded durable cross-departmental collaborations and shared infrastructure, and built an inclusive community of practice, contributing to a culture of collaboration and research excellence at Birmingham.

SELECTED PUBLICATIONS

Author name shown in bold. Up-to-date citation metrics are available on the linked Google Scholar profile.

Highlighted work

Abdlkarim, D., Di Luca, M., Aves, P., Maaroufi, M., Yeo, S.-H., Miall, R. C., Holland, P., & Galea, J. M. (2024). A methodological framework to assess the accuracy of virtual reality hand-tracking systems: A case study with the Meta Quest 2. *Behavior Research Methods, 56*(2), 1052–1063. *First author; a widely cited methodological contribution now used across VR, HCI and motor-science research.*

Watkins, F., **Abdlkarim, D.**, Winter, B., & Thompson, R. L. (2024). Viewing angle matters in British Sign Language processing. *Scientific Reports, 14*(1), 1043. *Cross-departmental sign-language collaboration applying kinematic methods.*

Bhatia, A., Mughrabi, M. H., **Abdlkarim, D.**, Di Luca, M., González-Franco, M., Ahuja, K., & Seifi, H. (2025). Text Entry for XR Trove (TEXT): Collecting and analyzing techniques for text input in XR. *CHI '25: Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems. Top-tier HCI venue (ACM CHI).*

Abdlkarim, D., Mukherjee, D., Giunchi, D., Di Luca, M., & Ofek, E. (2026). Belt and whistles: Adding lower-body collision awareness for MR experiences. *CHI '26: Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems.* *First author; ACM CHI.*

Journal articles

Abdlkarim, D., Di Luca, M., Aves, P., Maaroufi, M., Yeo, S.-H., Miall, R. C., Holland, P., & Galea, J. M. (2024). A methodological framework to assess the accuracy of virtual reality hand-tracking systems: A case study with the Meta Quest 2. *Behavior Research Methods, 56*(2), 1052–1063.

Watkins, F., **Abdlkarim, D.**, Winter, B., & Thompson, R. L. (2024). Viewing angle matters in British Sign Language processing. *Scientific Reports, 14*(1), 1043.

Schuster, B. A., Sowden, S. L., **Abdlkarim, D.**, Wing, A. M., & Cook, J. L. (2019). Acting is not the same as feeling: Emotion expression in gait is different for posed and induced emotions. *Frontiers in Human Neuroscience.*

Peer-reviewed conference papers

Abdlkarim, D., Mukherjee, D., Giunchi, D., Di Luca, M., & Ofek, E. (2026). Belt and whistles: Adding lower-body collision awareness for MR experiences. *ACM CHI Conference on Human Factors in Computing Systems (CHI '26)*, Barcelona.

Hauck, S., **Abdlkarim, D.**, Dudley, J., Kristensson, P. O., Ofek, E., & Grubert, J. (2025). ReachVox: Clutter-free reachability visualization for robot motion planning in virtual reality. *IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2025)*, pp. 1471–1478.

Bhatia, A., Mughrabi, M. H., **Abdlkarim, D.**, Di Luca, M., González-Franco, M., Ahuja, K., & Seifi, H. (2025). Text Entry for XR Trove (TEXT): Collecting and analyzing techniques for text input in XR. *ACM CHI Conference on Human Factors in Computing Systems (CHI '25)*, Yokohama.

González-Franco, M., **Abdlkarim, D.**, Bhatia, A., MacGregor, S., Fotso-Puepi, J. A., Gonzalez, E. J., Seifi, H., Di Luca, M., & Ahuja, K. (2024). Hovering over the key to text input in XR. *IEEE International Symposium on Emerging Metaverse (ISEMV 2024)*, pp. 13–16.

Ortenzi, V., Filipovica, M., **Abdlkarim, D.**, Pardi, T., Takahashi, C., Wing, A. M., Di Luca, M., & Kuchenbecker, K. J. (2022). Robot, pass me the tool: Handle visibility facilitates task-oriented handovers. *ACM/IEEE International Conference on Human-Robot Interaction (HRI '22)*, pp. 256–264.

Abdlkarim, D., Ortenzi, V., Pardi, T., Filipovica, M., Wing, A. M., Kuchenbecker, K. J., & Di Luca, M. (2021). PrendoSim: Proxy-hand-based robot grasp generator. *18th International Conference on Informatics in Control, Automation and Robotics (ICINCO 2021)*, pp. 60–68.

Conference abstracts & posters

Wei, M. X., **Abdlkarim, D.**, Yeo, S.-H., & Thompson, R. L. (2025). Investigating linguistic and motor influences in second sign-language production. *10th Conference of the International Society for Gesture Studies*, Nijmegen, Netherlands.

Wei, X., Thompson, R. L., **Abdlkarim, D.**, & Yeo, S.-H. (2025). Exploring linguistic and motor influences in signed second-language acquisition (poster). *15th Theoretical Issues in Sign Language Research Conference*, Addis Ababa, Ethiopia.

Preprints & research software

Tomczak, M., Li, M. S., Bradbury, A., Elliott, M., Stables, R., Witek, M., Goodman, T., **Abdlkarim, D.**, Di Luca, M., Wing, A. M., & Hockman, J. (2022). Annotation of soft onsets in string ensemble recordings. *arXiv preprint.*

Abdlkarim, D., Filipovica, M., Pardi, T., Ortenzi, V., Wing, A. M., Kuchenbecker, K. J., & Di Luca, M. (2021). PrendoSim, proxy-hand-based robot grasp generator (research software). *prendosim.github.io*.

EDUCATION

PhD, Computational Neuroscience 2016 – 2022
University of Birmingham, UK. EPSRC-funded. Sensorimotor control and neuroplasticity through immersive virtual-reality training.

MSc, Computational Neuroscience & Cognitive Robotics (CNCR) 2013 – 2014
University of Birmingham, UK. Awarded with Distinction.

BSc, Neuroscience 2009 – 2012
University of Manchester, UK. Final-year project on neuronal information processing in the brain.

RESEARCH & PROFESSIONAL EXPERIENCE

ARIA-funded Postdoctoral Research Fellow, University of Birmingham, UK 2025 – present
Non-invasive BCI for decoding naturalistic human movement: real-time neural signal processing, motion capture and neurophysiology to generate scientific results from first principles.

EPSRC-funded Postdoctoral Research Fellow, University of Birmingham, UK 2022 – 2025
Immersive XR and motion-capture research, including methodological work on VR hand-tracking accuracy and contributions to the Augmented Reality for Music Ensemble (ARME) programme.

Research Scientist, Meta (Reality Labs) Jun 2021 – Jun 2022
Addressed future directions in research design for novel wrist-based haptic feedback, building complex immersive simulations to test user performance and behaviour under real-time haptic stimulation.

Research Intern, Meta (Reality Labs) Jun 2019 – Dec 2019
Developed a series of experiments on human embodiment and perception for novel hand-tracking technologies, studying user behaviour in naturalistic contexts.

ENTREPRENEURSHIP

Co-Founder & Director, Motion Dynamics Ltd (Company No. 15797570) Jun 2024 – present
Building a mobile, computer-vision sports-analytics tool that combines computer vision with domain knowledge to deliver engaging performance feedback, in partnership with the School of Sport, Exercise & Rehabilitation Sciences and the School of Computer Science. Currently applying for Innovate UK funding following internal funding and University support. motiondynamics.ai

Co-Founder, Obi Robotics Ltd
Developing an all-in-one wireless data glove for applications in VR, health and robotics.

FUNDING & AWARDS

- **Innovate UK ICURe Explore & Exploit**, technology transfer (**£50K**), 2024–2025. One of 16 candidates accepted UK-wide; as entrepreneurial lead, building industry relationships and preparing a university spin-out in support of Birmingham's business and industry engagement strategy.
- **Google collaborative grant** (**£15K**), 2024–2025, developing and testing new text-entry techniques in extended reality (XR); contributed to an accepted ACM CHI 2025 paper.
- **BBSRC IAA, Adaptive Touch Testing System accelerator** (**£5K**), co-applicant, 2022–present, a mobile, accessible plug-and-play touch-testing system to automate and speed up clinical peripheral-neuropathy assessment of the hand and fingers.

REFERENCES

Prof. Katja Kornysheva, Centre for Human Brain Health; ARIA-grant lead and mentor.
k.kornysheva@bham.ac.uk

Prof. Massimiliano Di Luca, Principal Investigator, ARME project. m.diluca@bham.ac.uk

Prof. Chris Miall, PhD supervisor; Emeritus Professor of Motor Neuroscience. r.c.miall@bham.ac.uk